Resource-to-Need Alignment Matters: How Equity-Targeted Resources Shape Student Outcomes in CPS

Andi Fang Minyang Xu

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Group Information

- **Group Members:** Andi Fang, Minyang Xu
- **Lecture Section:** Tue/Thu 9:30am - 10:50am, Section 3
- **GitHub Usernames:** andifang-123(Andi Fang), clorisaaaaamelody(Minyang Xu)
- **Repository:** andifang-123/adf-myx-gp

1. Research Motivation & Question

Our project investigates how school resources and structural inequalities relate to student performance in Chicago Public Schools (CPS). Motivated by personal observations—one teammate worked with CPS through an internship, and the other observed strong on-campus facilities at a nearby school—we ask a single guiding question: **Are school resources aligned with student need, and how does this alignment relate to standardized academic outcomes (ELA and Math)?**

2. Approach & Methodology

To move beyond simple per-student funding metrics, we developed two composite indices.

The Resource Index (RI) – Measuring Capacity

RI combines physical (**Hard**) and institutional (**Soft**) resources, weighted equally.

- **Hard Resource Index:** We used **Space Utilization (SU)** as a proxy for physical capacity because facility "stock value" data were largely missing. We defined an efficiency center at (c = 0.85) with an overcrowding penalty factor gamma = 1.7, which is more harmful to learning.

  Distance to efficient use is defined as:

$$d(SU) = \begin{cases} c - SU, & SU < 0.7 \\ |SU - c|, & 0.7 \leq SU \leq 1.0 \\ \gamma(SU - c), & SU > 1.0 \end{cases}$$

The hard-resource score is:

$$Hard = -z(d(SU)).$$

- **Soft Resource Index:** We adopted the **5Essentials (5E)** framework—a research-based annual school improvement survey that measures five organizational conditions linked to student outcomes. It captures school climate and capacity across: Ambitious Instruction (AI), Effective Leaders (EL), Collaborative Teachers (CT), Involved Families (IF), and Supportive Environment (SE). Each domain is standardized via z-scores and averaged:

$$Soft = \frac{1}{5}\left(z_{AI} + z_{EL} + z_{CT} + z_{IF} + z_{SE}\right).$$

- **Final Resource Index (RI):**

$$RI = 0.5\,Hard + 0.5\,Soft.$$

Interpretation: Higher **Resource Index (RI)** indicates stronger overall school capacity—both physical capacity/availability and stronger organizational (5E) conditions.

- **The Opportunity Constraint Index (OI) – Measuring Need**

OI quantifies student need by averaging z-scores across four domains: Economic Disadvantage %, English Learners %, Disabilities %, and Race Concentration (measured via a Simpson-based concentration index):

$$Simpson = \sum_r p_r^2.$$

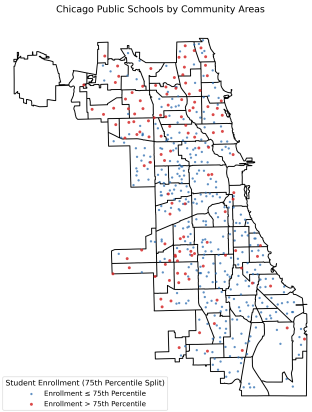
Higher **OI** indicates greater opportunity constraints (and thus greater resource need to achieve comparable outcomes).

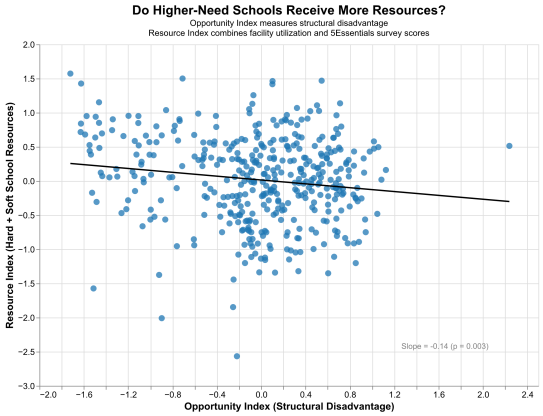
- **Weaknesses**
  - **Resource measurement limits:** Hard resources are proxied by space utilization; we lack granular data on direct spending or staffing counts.
  - **Index construction assumptions:** OI assumes equal weights across components, though economic disadvantage might plausibly deserve a higher weight.
  - **Interpretation scope:** Our results are descriptive and do not establish causal effects of resources on outcomes.

3. Static Visualizations

Our static plots provide the baseline for our structural analysis:

- **Spatial Distribution of Public Schools:** This map shows that schools are unevenly distributed. Schools with higher enrollment (above the 0.75 quantile) are visibly clustered in the North and West community areas compared to southern and central areas. *(left figure)*
- **Do Higher-Need Schools Receive More Resources? (RI vs OI):** The RI–OI scatter plot provides a baseline diagnostic of resource–need alignment by comparing the **Resource Index (RI)** against the **Opportunity Constraint Index (OI)** across schools. *(right figure)*





4. Streamlit Applications

We developed two interactive apps to explore these dynamics in depth:

- **Subgroup Distribution App:** (https://adf-myx-gp-structural-disadvantage.streamlit.app/)

This tool allows users to toggle between ELA and Math results to see if performance differs systematically across disadvantage groups. It visualizes that structural barriers are strongly reflected in proficiency distributions.

- **Spatial & Scatter Explorer** (https://adf-myx-gp-sptial-scatter.streamlit.app/):
  - **Spatial View:** Maps community-average **Resource Index (RI)** and **Opportunity Index (OI)** scores, listing the top and bottom 10 communities. This enables cross-comparisons to identify mismatch areas (e.g., **low RI but high OI**, or **high RI but low OI**) for further review.
  - **Regression View:** Links academic outcomes (**ELA/Math proficiency**) to RI and OI. Across the four scatter plots, the fitted lines summarize the substantive pattern: **higher RI/lower OI is generally associated with higher proficiency**, highlighting how resources and constraints relate to student outcomes.

5. Substantive Results & Conclusion

Our primary finding is **resource–need misalignment**: the relationship between **Resource Index (RI)** and **Opportunity Index (OI)** is flat or negative, whereas a need-aligned system would be **non-negative on average** (higher constraints should not systematically coincide with lower resources). We do not claim a single "correct" numeric slope, but under our framework the expected **direction** is non-negative (potentially nonlinear), so a flat/negative pattern is substantively meaningful as a diagnostic of potential misallocation.

Policy Recommendation: Use **OI** and **RI** jointly to target and re-balance support — prioritize **high-need, low-resource** communities for additional investment, review **low-need, high-resource** areas for potential re-targeting, and move toward a need-aligned allocation rule where OI helps guide investments and track whether gaps narrow over time.